An upper bound $f(6) \leq 20$ for almost-equidistant sets in $\mathbb{R}^6$, and a reduction of $f(6) \leq 19$ to thirty-four graphs

A computer-assisted frontier argument over a certified level-19 ledger

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Background

A finite set of points in $\mathbb{R}^d$ is *almost equidistant* if among any three of its points, some two are at unit distance. Let $f(d)$ denote the largest size of an almost-equidistant set in $\mathbb{R}^d$. Balko, Pór, Scheucher, Swanepoel and Valtr [1] established

$$18 \;\leq\; f(6) \;\leq\; 26,$$

the lower bound from the Larman–Rogers half-cube with two poles, the upper bound from the absence of admissible abstract candidates on 27 vertices. The values $f(4) = 12$ and $f(5) = 16$ were determined in the accompanying notes of this repository. This note proves, subject to the certificate ledger stated explicitly below,

$$f(6) \;\leq\; 20,$$

and reduces $f(6) \leq 19$ to the non-realizability of thirty-four explicit 20-vertex graphs, which are under interval certification at the time of writing.

Framework

For an almost-equidistant set $X \subset \mathbb{R}^6$ with unit-distance graph G: the complement $H = \overline{G}$ is triangle-free; G contains no K_8 (at most seven points are pairwise at unit distance in $\mathbb{R}^6$); and G contains no $K_{1,3,3,3}$ (Lemma 11 of [1], even dimensions). Deleting unit edges only weakens these constraints, so every n-point almost-equidistant set yields, after completing H to a maximal triangle-free graph, a *level-n candidate*: a graph on n vertices whose complement is maximal triangle-free, with no K_8 and no $K_{1,3,3,3}$, all of whose edges are realized at unit distance by the n points. The repository's verified level-19 corpus contains exactly 3,971,787 such candidates [1, 2].

The level-19 ledger

Write $\mathcal{R}$ for the set of 644 level-19 candidates (19 containing K_7, 625 with clique number six) listed in `residue644.txt`. The ledger assumption is:

> **Ledger.** Every level-19 candidate outside $\mathcal{R}$ is non-realizable by 19 distinct points in $\mathbb{R}^6$.

Provenance of the ledger, in decreasing order of independent scrutiny:

1. 3,855,094 candidates are rejected by exact integer graph filters (link bounds via $f(5) = 16$, $f(4) = 12$, $f(3) = 10$ and a circle argument; K_7 facet-reflection coincidences; a K_7 defect-support CSP; the K_7 disjoint-edge bounded-cover rule with its tight-cover matching refinement; a K_6 two-light-ray CSP). These rules were developed on the `codex/dimension6`

track, their mathematical derivations were verified by hand, and the entire rule set was *independently reimplemented from the theory document alone* (`indep_profile6.c`); the two implementations agree exactly on all nine per-rule corpus statistics, including the residue split 115,587 / 1,106.

2. 2,451 of the remaining candidates carry interval-arithmetic non-realizability certificates from this repository's certified engine (`ckernel6.c`, quota + mean-value enclosures, IEEE-754 outward rounding). A further 3,053,029 candidates in class 1 carry such certificates as well, redundantly.

3. 114,242 candidates are rejected only by the later exact layers of the `codex/dimension6` track (positive-semidefinite Z-matrix chains, Hall layers, support/pentad conjunctions, tetrad identities, empty-support budgets, virtual-K_7 closure). Each layer ships an independent checker that reconstructs its inputs and replays its certificates (all PASS); an independent reimplementation of these layers, rule by rule, is in progress but not complete. This class is the conditional part of the ledger.

The extension theorem

Theorem 1 (extension parametrization)**.** *Let R be a level-n candidate with complement H_0, and let C be a level-$(n{+}1)$ candidate on $V(R) \cup \{w\}$ such that $C - w \supseteq R$ (edge containment on the same vertex set). Let N be the set of non-neighbours of w in C. Then*

$$C[V(R)] \;=\; R + H_0[N], \qquad N(w) = V(R) \setminus N,$$

where $H_0[N]$ is the set of H_0-edges inside N; moreover $|N| \le 7$, and N dominates $V(R) \setminus N$ in H_0.

Proof. Write D for the set of edges of $C[V(R)]$ not in R; each is an H_0-edge. Let $H = \overline{C}$. (i) $D \subseteq N \times N$: for $\{u, v\} \in D$, the pair uv is a non-edge of H; since H is maximal triangle-free, adding it must create a triangle through some common H-neighbour z. No $z \in V(R)$ works: $H - w \subseteq H_0$ and a common H_0-neighbour of the H_0-edge uv would be a triangle in the triangle-free H_0. Hence $z = w$, i.e. $u, v \in N$. (ii) $H_0[N] \subseteq D$: an H_0-edge inside N surviving in H would form a triangle with w. Thus $D = H_0[N]$ exactly. $|N| \le 7$ because N becomes a clique of C (all pairs inside N are C-edges after the D-flips, since $H[N]$ is empty), and C has no K_8. Domination is the maximality of H at the pairs wu, $u \notin N$: a triangle witness z must satisfy $z \in N$ and $zu \in H \subseteq H_0$. □

Conversely, given R and any $N \subseteq V(R)$, the displayed formulas define a graph whose candidacy is checked directly (complement triangle-free and maximal, no K_8, no $K_{1,3,3,3}$). The enumeration over all $|N| \le 7$ with these direct checks is therefore a *complete* enumeration of the level-$(n{+}1)$ candidates edge-containing a given set of level-n candidates at some deletion.

Lemma 1 (hereditary frontier)**.** *Assume the Ledger. If an n-point almost-equidistant set exists in $\mathbb{R}^6$ ($n \ge 20$), then every level-n candidate derived from it has the property that for every vertex u and* every *maximal triangle-free completion of $\overline{C - u}$, the complementary level-$(n-1)$ candidate is again derived from an $(n-1)$-point set; for $n - 1 = 19$ this forces membership in $\mathcal{R}$.*

Proof. If P realizes C (all C-edges unit), then $P \setminus u$ realizes every candidate obtained from $C - u$ by completion, since completion only removes required edges. A certified non-realizable completion is a contradiction; by the Ledger every completion at level 19 lies in $\mathcal{R}$. □

The computation

All tools, logs and outputs are in the repository; the enumerator is `ext6.c` (exact canonical forms: signature refinement, invariant trace pruning, verified-automorphism orbit pruning; every automorphism used for pruning is re-verified to preserve adjacency).

Controls. (a) Extensions of the full verified level-14 corpus (1,052 graphs) reproduce the independently generated level-15 corpus *exactly* as a set of isomorphism classes (3,969; canonical set equality). (b) Extensions of level 15 give 18,917 distinct classes, matching the documented level-16 count. (c) Canonical-form invariance under random relabelings over both corpora. (d) The complete pipeline was re-implemented independently in Python (own subgraph tests, own isomorphism decision via profile buckets and backtracking, own completion enumeration); it reproduces both control counts exactly, including the raw multiplicities 250,339 and 962,217.

Level 20. The 644 residue graphs admit 223,847 valid extensions forming 117,290 distinct level-20 candidates. For each, every one of its twenty single-vertex deletions was re-completed greedily; 117,256 candidates have a deletion whose completion is a certified level-19 graph, killing them by the Lemma. For each of the remaining 34, *all* maximal triangle-free completions of all deletions were enumerated: every completion lies in $\mathcal{R}$. The level-20 frontier is exactly these 34 graphs (`frontier20_true.txt`; all have clique number six and 130–140 edges against 99 degrees of freedom).

Level 21. The 34 frontier graphs admit 3,602 distinct level-21 extensions. Every single one has a deletion whose greedy re-completion falls outside the frontier; since membership in the level-20 frontier is (by the Theorem's completeness) a necessary condition for a level-20 candidate to be derived from a 20-point set, each such completion is underivable, and the level-21 frontier is *empty*.

Result. *Assume the Ledger. Then no* 21*-point almost-equidistant set exists in* $\mathbb{R}^6$*; combined with heredity,* $f(6) \leq 20$*. Moreover* $f(6) \leq 19$ *if and only if none of the* 34 *frontier graphs is realizable by* 20 *distinct points with all listed edges at unit distance.*

A Levenberg–Marquardt screen (500 restarts per graph) puts every one of the 34 far from realizability (best squared-residual 0.22, typical 0.4–1.5, against 10^{-30} for realizable controls); their interval certification is running. One graph (index 32: a 13-regular, K_7-free, vertex-degree-uniform candidate with twenty K_6's) requires a placement stage not yet implemented in the engine (two-sphere vertices) and is flagged explicitly.

Reproduction

```
clang -O2 -o ext6 ext6.c
./ext6 canontest aeq_d6_n14.txt 8       # canonical-form self-test
./ext6 extend aeq_d6_n14.txt -          # control: 3,969 classes = n15 corpus
./ext6 extend aeq_d6_n15.txt -          # control: 18,917 classes
python extract_residue_v6.py            # residue644.txt, verified vs corpus
./ext6 extend residue644.txt - prov20.txt > ext20_raw.txt
./ext6 frontier ext20_raw.txt residue644.txt    # 117,256 killed, 34 survive
python escalate_frontier.py frontier20.txt residue644.txt   # all-completions
./ext6 extend frontier20_true.txt - > ext21_raw.txt
./ext6 frontier ext21_raw.txt frontier20_true.txt          # 0 survive
python verify_frontier_indep.py         # independent re-run of everything
clang -O2 -o indep_profile6 indep_profile6.c
./indep_profile6 aeq_d6_n19.txt v.bin   # exact-filter cross-check
```

References

[1] M. Balko, A. Pór, M. Scheucher, K. Swanepoel, P. Valtr, Almost-equidistant sets, *Graphs and Combinatorics* 36 (2020), 729–754.

[2] This repository: candidate corpora, certified interval engine, exact-filter implementations, verifiers, and logs (`STATUS.md` for the campaign record).